Infection Prevention and Control and Biomedical Waste Management

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1. Purpose

This document describes the infection prevention and waste management procedures at CareOps General Hospital. It follows public guidance from the World Health Organization, the United States Centers for Disease Control and Prevention and the Biomedical Waste Management Rules 2016 of India, and adds internal operational rules. It is used by the operations team to understand isolation bed needs, waste handling and infection control actions that affect patient flow.

2. Hand hygiene

The World Health Organization defines five moments for hand hygiene: before touching a patient, before a clean or aseptic procedure, after body fluid exposure risk, after touching a patient, and after touching patient surroundings. Alcohol-based hand rub is used when hands are not visibly dirty and the rub should take 20 to 30 seconds. Handwashing with soap and water is used when hands are visibly dirty or soiled with blood or body fluids and should take 40 to 60 seconds. The hospital audits hand hygiene compliance every month.

3. Standard precautions

Standard precautions apply to all patients at all times. They include hand hygiene, use of gloves, gowns, masks and eye protection according to the risk of contact with blood or body fluids, safe handling of sharps, cleaning and disinfection of equipment and surfaces, safe handling of linen, respiratory hygiene and cough etiquette, and safe waste disposal.

4. Transmission-based precautions

In addition to standard precautions, three categories are used. Contact precautions are used for infections spread by direct or indirect contact, for example MRSA or Clostridioides difficile, and need gloves and a gown and preferably a single room. Droplet precautions are used for infections spread by large respiratory droplets, for example influenza, and need a surgical mask and a single room where possible. Airborne precautions are used for infections that spread through small particles that stay in the air, for example tuberculosis, measles and chickenpox. They need an N95 respirator for staff and a negative pressure airborne infection isolation room. The Isolation Unit has 20 beds and 4 of them are negative pressure rooms.

5. Isolation and bed management

The Infection Control Officer decides when a patient needs isolation. When isolation beds are almost full, patients with contact precautions may be placed in single rooms in general wards after approval. Airborne isolation patients may only be placed in negative pressure rooms. After an isolation patient leaves, the bed stays in isolation hold until terminal cleaning of 60 to 90 minutes is complete. If all 4 negative pressure rooms are occupied, the Duty Manager is informed immediately.

6. Outbreak response

Two or more linked cases of the same infection in one ward within 7 days are reported to the Infection Control Officer as a possible outbreak. The officer visits the ward, reviews cases, orders cohorting of patients if needed, restricts admissions to the ward if necessary and informs the Medical Superintendent. Outbreaks that need notification are reported to the public health authority.

7. Biomedical waste colour coding

The Biomedical Waste Management Rules 2016 of India set colour-coded containers. Yellow bags are for human anatomical waste, soiled waste such as dressings and items contaminated with blood, expired or discarded medicines and chemical waste. Red bags are for contaminated recyclable plastic waste such as tubing, catheters, intravenous sets and gloves. White translucent puncture-proof, leak-proof and tamper-proof containers are for sharps such as needles, syringes with fixed needles and scalpel blades. Blue cardboard boxes with a blue marking are for broken or discarded glass and metallic body implants. General non-infectious waste such as paper and food packaging goes to the general waste bins.

8. Waste handling rules

Under the Rules untreated biomedical waste must not be stored for more than 48 hours. Waste is segregated at the point of generation and is never mixed. Bags are closed when three quarters full. Waste is moved on designated routes and trolleys to the central storage room and is collected by an authorised common biomedical waste treatment facility. A barcode system is used on bags and containers so that waste can be tracked. Staff who handle waste are vaccinated against hepatitis B and use protective equipment. Needle stick injuries are reported at once and the injured staff member receives post-exposure assessment within 2 hours.

9. Environmental cleaning and linen

High-touch surfaces in patient areas are cleaned at least twice daily and in the ED and ICU at least three times daily. Spills of blood or body fluids are cleaned immediately using a disinfectant. Infected linen is placed in leak-proof bags and sent to the laundry separately.